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Devops B2

Lab Exercise 19- ClusterIP in Service in Kubernetes

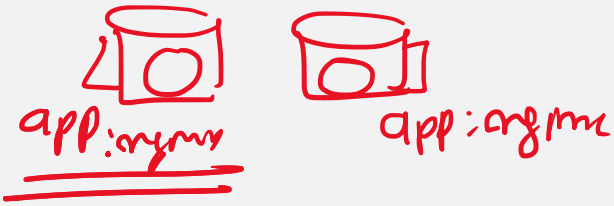
A ClusterIP is the default type of service in Kubernetes. It provides a stable internal IP address for the service, which is accessible only within the cluster. This is useful for internal communication between services within the cluster.

Step 1: Create Deployment (Correct Labels)

Create file:

nginx-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx   # MUST match template labels
  template:
    metadata:
      labels:
        app: nginx  # MUST match service selector
    spec:
      containers:
      - name: nginx
```



```
    image: nginx
    ports:
    - containerPort: 80
```

Save file.

```
kubectl apply -f nginx-deployment.yaml
```

```
C:\Users\prati>notepad nginx-deployment.yaml

C:\Users\prati>kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-app created
```

Verify:

```
kubectl get pods --show-labels
```

```
C:\Users\prati>kubectl get pods --show-labels
NAME                          READY   STATUS              RESTARTS      AGE
    LABELS
devops-app-6fc4b9fb96-999vw   1/1     Running             4 (27m ago)   3d17
h   app=devops-app,pod-template-hash=6fc4b9fb96
devops-app-6fc4b9fb96-p9tb4   1/1     Running             4 (27m ago)   3d17
h   app=devops-app,pod-template-hash=6fc4b9fb96
nginx-app-75fdcbbc74-9lszh    0/1     ContainerCreating   0             6s
    app=nginx,pod-template-hash=75fdcbbc74
nginx-app-75fdcbbc74-v7c27    1/1     Running             0             6s
    app=nginx,pod-template-hash=75fdcbbc74
```

You should see:

```
app=nginx
```

Step 2: Create ClusterIP Service (Matching Selector)

Create file:

nginx-service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx   # MUST match pod label
  ports:
  - port: 80
    targetPort: 80
```

Apply:

```
kubectl apply -f nginx-service.yaml
```

```
C:\Users\prati>notepad nginx-service.yaml

C:\Users\prati>kubectl apply -f nginx-service.yaml
service/nginx-service created
```

Step 3: Verify It Is Working

Check Service

```
kubectl get svc
```

```
C:\Users\prati>kubectl get svc
NAME             TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        A
GE
devops-service   NodePort    10.109.111.177   <none>        80:30008/TCP   3
d17h
kubernetes       ClusterIP   10.96.0.1        <none>        443/TCP        3
d17h
nginx-service    ClusterIP   10.100.252.193   <none>        80/TCP         4
s
```

You should see:

```
TYPE       CLUSTER-IP
ClusterIP  10.x.x.x
```

MOST IMPORTANT CHECK

```
kubectl get endpoints nginx-service
```

```
C:\Users\prati>kubectl get endpoints nginx-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 Endpo
intSlice
NAME            ENDPOINTS                       AGE
nginx-service   10.244.0.20:80,10.244.0.21:80   14s
```

Now you should see something like:

```
10.244.x.x:80, 10.244.x.x:80
```

Step 4: Test From Inside Cluster

Run test pod:

```
kubectl run test-pod --image=busybox --rm -it – sh
```

Inside pod:

```
wget -qO- http://nginx-service
```

You should see:

```
Welcome to nginx!
```

Exit:

```
Exit
```

```
C:\Users\prati>kubectl run test-pod --image=busybox --rm -it -- sh
All commands and output from this session will be recorded in container logs, including cre
dentials and sensitive information passed through the command prompt.
If you don't see a command prompt, try pressing enter.
/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit
Session ended, resume using 'kubectl attach test-pod -c test-pod -i -t' command when the po
d is running
pod "test-pod" deleted from default namespace
```

Step 5: Test From Outside (It Will Fail)

From your local machine:

```
curl http://<CLUSTER-IP>
```

```
C:\Users\prati>kubectl get svc
NAME             TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
devops-service   NodePort    10.109.111.177   <none>        80:30008/TCP   3d17h
kubernetes       ClusterIP   10.96.0.1        <none>        443/TCP        3d17h
nginx-service    ClusterIP   10.100.252.193   <none>        80/TCP         63s

C:\Users\prati>curl http://<CLUSTER-IP>
The syntax of the command is incorrect.

C:\Users\prati>kubectl get pods
NAME                          READY   STATUS    RESTARTS      AGE
devops-app-6fc4b9fb96-999vw   1/1     Running   4 (29m ago)   3d17h
devops-app-6fc4b9fb96-p9tb4   1/1     Running   4 (29m ago)   3d17h
nginx-app-75fdcbbc74-9lszh    1/1     Running   0             2m7s
nginx-app-75fdcbbc74-v7c27    1/1     Running   0             2m7s

C:\Users\prati>kubectl delete deployment nginx-app
deployment.apps "nginx-app" deleted from default namespace

C:\Users\prati>kubectl delete service nginx-service
```

It will NOT work.

That proves:

ClusterIP works only inside cluster.